METHOD SUMMARY

# Chloride in Water by Ion Chromatography

Reference method: EPA Method 300.0, Determination of Inorganic Anions by Ion Chromatography, Revision 2.1 (1993)

**Accredited method.** Riverside Analytical is accredited by the Minnesota Department of Health Environmental Laboratory Accreditation Program (MNELAP) for chloride in non-potable water by EPA 300.0.

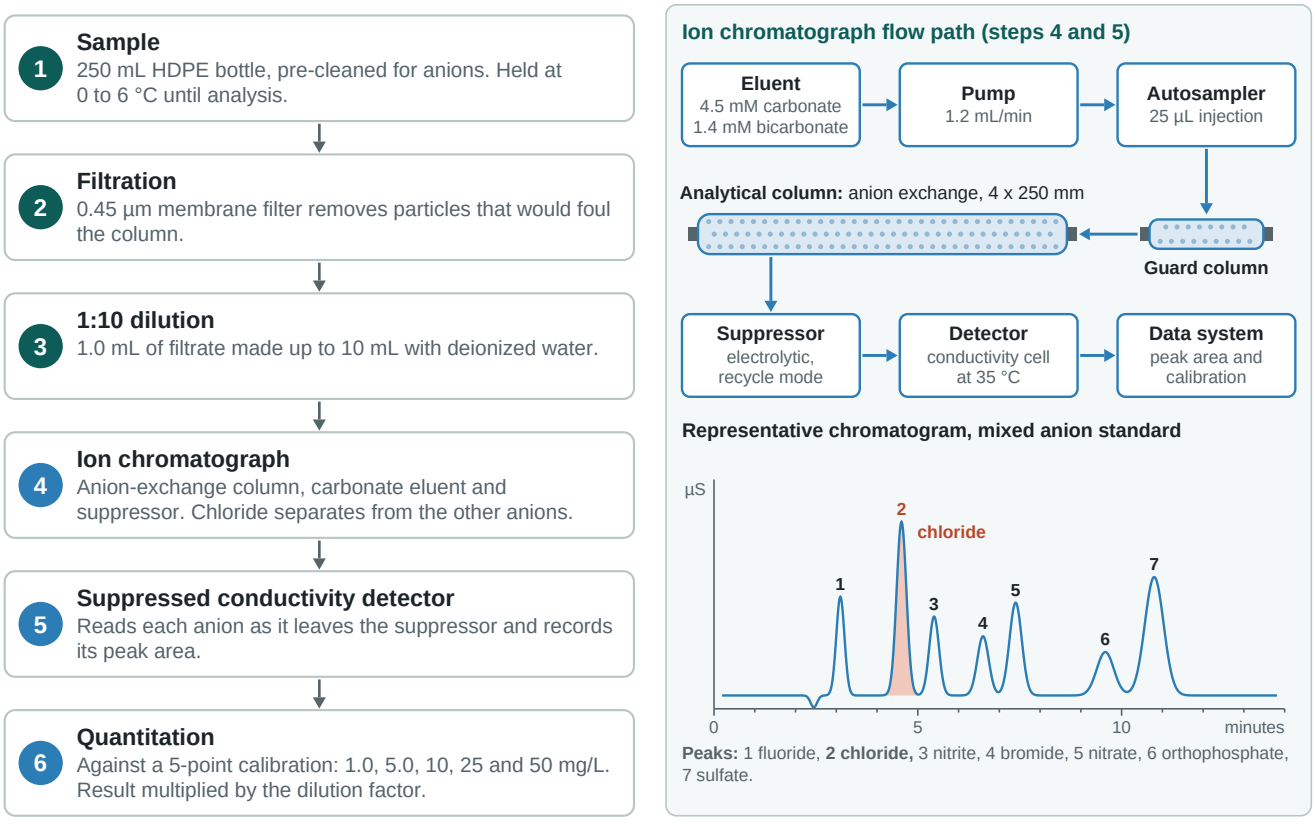
## SCOPE

Chloride in surface water, stormwater, groundwater and other non-potable water. The same run also separates fluoride, nitrite, bromide, nitrate, orthophosphate and sulfate, which are reported only when requested. This summary covers chloride.

## PRINCIPLE

A filtered portion of each sample is diluted 1:10 and injected into the ion chromatograph (Figure 1). The anions separate on an anion-exchange column in a sodium carbonate and sodium bicarbonate eluent. The suppressor then exchanges the sodium in the eluent for hydrogen, turning the eluent into weakly conducting carbonic acid, so the conductivity detector reads each anion against a low background. Chloride is identified by its retention time and measured by peak area against a five-point calibration.

**Figure 1.** Sample workflow, and the flow path inside the ion chromatograph




## CONDITIONS

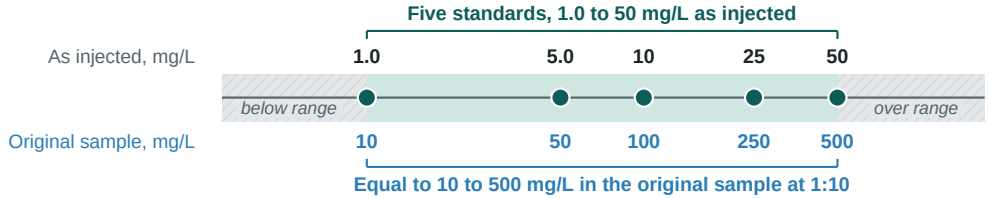
Anion-exchange guard and analytical columns at 30 °C; eluent 4.5 mM sodium carbonate and 1.4 mM sodium bicarbonate at 1.2 mL/min; 25 µL injection; electrolytic suppressor in recycle mode; run time 15 minutes per injection.

## CALIBRATION

Five standards, 1.0, 5.0, 10, 25 and 50 mg/L chloride, are prepared from a certified stock solution and run at the start of each analytical run. Peak area is fitted to concentration by linear least squares, and the correlation coefficient (r) must be 0.995 or better before any sample is run.

**Calibrated range:** 1.0 to 50 mg/L as injected, equal to 10 to 500 mg/L in the original sample at 1:10. A sample reading above the top standard is diluted further and rerun. A sample reading below the lowest standard is rerun at a lower dilution. Results are not extrapolated outside the calibrated range.

**Figure 2.** Calibrated range at the standard 1:10 dilution (log scale)




## QUALITY CONTROL

Each analytical batch holds up to 20 samples and carries the checks below. A batch that fails a check is rerun, or the affected results are qualified on the report.

| **QC check** (EPA 300.0 term) | **How often** | **Acceptance limit** | **WO-26-10418,** batch IC-261016-2 |
|---|---|---|---|
| **Initial calibration,** five standards, 1.0 to 50 mg/L | Start of each analytical run | r of 0.995 or better | r = 0.9997 |
| **Continuing calibration check** (instrument performance check) | Every 10 injections and at the end of the run | 90 to 110% of the true value | All within limits |
| **Method blank** (laboratory reagent blank) | Each batch | Below the reporting limit | <0.5 mg/L |
| **Laboratory control sample** (laboratory fortified blank) | Each batch | 90 to 110% recovery | 101% recovery |
| **Matrix spike** (laboratory fortified sample matrix) | Each batch | 80 to 120% recovery | 98% recovery, spiked on COR-1 |
| **Sample duplicate** | Each batch | RPD 10% or less | RPD 2.1%, duplicate of COR-1 |

**Batch column:** results for work order WO-26-10418, analytical batch IC-261016-2, analyzed 2026-10-16, as reported on its certificate of analysis. RPD is the relative percent difference between duplicate results.

## SAMPLE HANDLING

| | |
|---|---|
| **Container** | 250 mL HDPE bottle, pre-cleaned for anions, supplied by the laboratory. |
| **Preservation** | No chemical preservative. Keep at 0 to 6 °C from collection to delivery. |
| **Holding time** | 28 days from collection. |
| **At receipt** | Cooler temperature, custody seal and bottle condition are checked and recorded. |

## INTERFERENCES

Anions that elute close to chloride, or a very large neighboring peak, can affect resolution; further dilution resolves most cases. Filtration at 0.45 µm protects the column.

## REPORTING

Results are reported in mg/L chloride to three significant figures, corrected for dilution. The certificate of analysis lists each sample's dilution, its reporting limit and the batch QC.

## REFERENCE

U.S. EPA Method 300.0, Determination of Inorganic Anions by Ion Chromatography, Revision 2.1 (August 1993). Approved for chloride under 40 CFR Part 136.

**Document control.** This summary reflects SOP RA-IC-300, Revision 7, effective 2025-11-03, which supersedes Revision 6. Riverside Analytical Quality Assurance reviews the SOP every year. Issued 2026-10-20 with work order WO-26-10418.